

Junqiao Fan

Curriculum Vitae

Ph.D. Candidate (thesis submitted) | Nanyang Technological University, Singapore
School of Electrical and Electronic Engineering | Expected graduation: December 2026

fanj0019@e.ntu.edu.sg | Homepage | Google Scholar | GitHub | LinkedIn

Open to postdoctoral opportunities.

Research Interests

Multimodal human perception: Privacy-preserving mmWave radar sensing; human motion capture and human pose/mesh estimation.

Generative AI: Human motion generation and modeling; diffusion and flow models.

Robotics: Robot learning; multimodal human-robot interaction; human-centered embodied intelligence.

Education & Research Experience

Nanyang Technological University, Singapore 2022–Present

Ph.D., School of Electrical and Electronic Engineering (thesis submitted).

Supervisor: Prof. Lihua Xie; Co-supervisor: Prof. Jianfei Yang.

The Hong Kong Polytechnic University, Hong Kong SAR 2018–2022

B.Eng. in Computer Science; minor in Mathematics.

Selected Publications

- mmHRI: Towards Privacy-Preserving Human-Robot Interaction with Millimeter-Wave Radar**
Junqiao Fan, Yuxuan Hu, Bofan Lyu, Yanshuo Lu, Pengfei Liu, Jiarui Zhang, Fangqiang Ding, Lihua Xie, Gen Li, Jianfei Yang
arXiv:2609.34220, 2026 (preprint). [Paper] [Project]
- GIVE: Grounding Human Gestures in Vision-Language-Action Models**
Pengfei Liu, Gen Li, Junqiao Fan, Boyu Ma, Jindou Jia, Yang Xiao, Jianfei Yang
arXiv:2606.13435, 2026 (preprint). [Paper] [Project]
- M4Human: A Large-Scale Multimodal mmWave Radar Benchmark for Human Mesh Reconstruction**
Junqiao Fan, Yunjiao Zhou, Yizhuo Yang, Xinyuan Cui, Jiarui Zhang, Lihua Xie, Jianfei Yang, Chris Xiaoxuan Lu, Fangqiang Ding
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. [Paper] [Project]
- mmPred: Radar-based Human Motion Prediction in the Dark**
Junqiao Fan, Haocong Rao, Jiarui Zhang, Jianfei Yang, Lihua Xie
AAAI Conference on Artificial Intelligence, 40(5):3777–3785, 2026. [Paper] [Project]
- Diffusion Model Is a Good Pose Estimator from 3D RF-Vision**
Junqiao Fan, Jianfei Yang, Yuecong Xu, Lihua Xie
European Conference on Computer Vision (ECCV), pp. 1–18, 2024. [Paper] [Project]
- Generative Dataset Distillation Using Min-Max Diffusion Model**
Junqiao Fan, Yunjiao Zhou, Min Chang Jordan Ren, Jianfei Yang
European Conference on Computer Vision Workshops, pp. 60–67, 2024. [Paper]
- Federated Learning Driven Secure Internet of Medical Things**
Junqiao Fan, Xuehe Wang, Yanxiang Guo, Xiping Hu, Bin Hu
IEEE Wireless Communications, 29(2):68–75, 2022. [Paper]

Honors & Awards

2nd Place, Generative Track 2024

First Dataset Distillation Challenge, ECCV 2024.

Academic Service

Journal Reviewer: TMC, etc.

Conference Reviewer: AAAI, ICLR, ICRA, etc.

Skills

Python, MoCap, ROS2.